

PH3202 (Intermediate EM)

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Assignment 9

Question 1

Derive the law:

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

from Maxwell's action:

$$S_{EM}[A_\mu] = \int d^4x \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + A_\nu J^\nu \right)$$

Question 2

Show that the integral

$$I_2 = \int_0^\pi d\theta \sin\theta e^{ikr \cos\theta} = \frac{2 \sin(kr)}{kr}$$

Question 3

Calculate the integral

$$I_3 = c \int_{-\infty}^{\infty} d\omega \frac{e^{-i\omega t}}{(\omega - ck)(\omega + ck)}$$